

Yosime Flood

Curriculum vitae

Mathematics · Humboldt-Universität zu Berlin
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Mathematical interests

Algebraic geometry, arithmetic geometry, and algebraic number theory.

Education

2024-present **M.Sc. Mathematics**, Humboldt-Universität zu Berlin
2020-2024 **B.Sc. Mathematics**, Technische Universität Berlin
Minor in Computer Science. Final grade: 1.5 (German scale; 1.0 is best).

Thesis and research experience

2024 **Algebraic Curves and the Riemann-Roch Theorem**
Bachelor's thesis, TU Berlin. Expository study based on Fulton's *Algebraic Curves*, including nonsingular models, divisors, and the Riemann-Roch theorem.

2023 **X-Student Research Group**, Geometry AG, TU Berlin
Investigated combinatorial solutions to discrete integrable systems under the supervision of Dr. Niklas Affolter.

Seminar talks and write-ups

Feb. 2025 **Regulär Lokale Ringe II**
Commutative algebra: regular local rings and global dimension. In German.

Mar. 2022 **Irreduzible Wurzelsysteme**
Lie theory: irreducible root systems and Weyl groups. In German.

Teaching and mathematical service

2022-2026 **Teaching Assistant**, Technische Universität Berlin
Numerical Mathematics (Oct. 2026)
Stochastics for Computer Science (Mar. 2026)
Early Bird: Analysis and Linear Algebra for Engineering Freshmen (Aug. 2025)
Analysis and Linear Algebra I for Engineers (Mar. 2025)
Abstract Algebra I (Oct. 2024 and Oct. 2023)
Analysis III for Mathematicians (Mar. 2024)
Analysis I for Mathematicians (Mar. 2023)
Analysis and Linear Algebra for Engineers (Oct. 2022)

2025 **Jury Member and Deputy Team Leader**, IMC, TU Berlin team
Coached and mentored students; evaluated competition solutions.

2021-2024 **German Mathematical Olympiad**, state-round grading committee
Graded solutions for the oldest age category.

Selected mathematical distinctions

2024 **Bronze medal**, International Mathematics Competition for University Students (IMC). Selected to represent TU Berlin.

2020 **Gold medal; team Grand First Prize**, International Tournament of Young Mathematicians (ITYM). Project on recursive sequences on rooted trees.

2024 **Top 25, advanced category**, Vojtěch Jarník International Mathematical Competition.

2019-2020 **Top 10 nationwide in the 2019 finals**, Bundeswettbewerb Mathematik. First Prizes in Rounds I and II in both 2019 and 2020.

Tools and languages

Python; LaTeX. German and English (bilingual); Korean (beginner).